1. What is the difference between Static IP and Dynamic IP?

Answer:

- Static IP: An IP address that remains the same until it is manually changed or reassigned.
- Dynamic IP: An IP address that is automatically assigned by DHCP and may change over time.

Static IP	**Dynamic IP**
Remains the same.	May change over time.
Configured manually or reserved.	Assigned automatically by DHCP.
Used for servers and network services.	Commonly used for home devices.
Requires more administration.	Easy to manage.

2. Why do home users usually receive Dynamic IP Addresses?

Answer:

Home users usually receive Dynamic IP addresses because they are easy to manage and configure automatically.

Internet Service Providers (ISPs) can reuse IP addresses when customers disconnect. Most home users do not need a permanent IP address.

3. What is NAT?

Answer:

NAT stands for Network Address Translation.

NAT is a process used by routers to translate Private IP addresses into a Public IP address when devices access the Internet.

It allows multiple devices to share a single Public IP address.

4. How does NAT help Private IP devices access the Internet?

Answer:

Private IP addresses cannot communicate directly over the public Internet.

When a device sends data to the Internet, the router uses NAT to replace the device's Private IP address with its Public IP address.

The router maintains a translation table to identify each connection.

When response data comes back, the router forwards it to the correct device using that table.

Example:

- Laptop: 192.168.1.10
- Mobile: 192.168.1.11
- Router Public IP: 49.205.100.25

Both the laptop and mobile can access the Internet using the same Public IP address through NAT.